



Database Management Systems

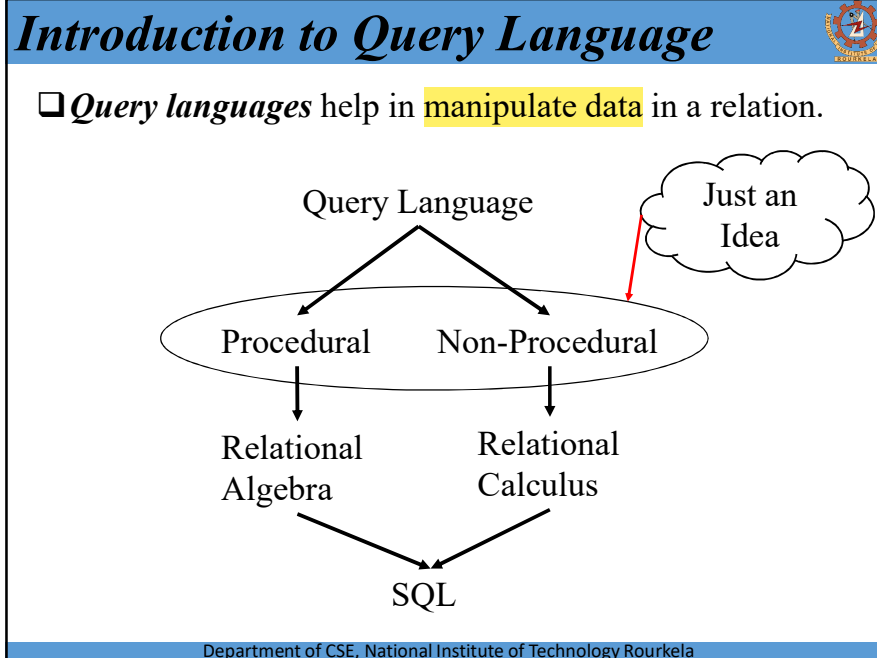
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Abstract Query Languages: Relational Algebra and Relational Calculus

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Introduction to Query Language

- ❑ *Query languages* help in manipulate data in a relation.
- ❑ The Query models (***Relational Algebra*** and ***Relational Calculus***) are abstract or theoretical.
- ❑ Relational Algebra and Relational Calculus (with only safe queries) are ***equivalent***.
- ❑ In practice, ***Relational model*** is implemented as RDBMS, and the query model is implemented as ***SQL***.
- ❑ A language is called relationally complete if it has capacity to express every query expressible through relational algebra. ***SQL is relationally complete.***

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Introduction to Query Language



- ❑ **Degree / Arity of a Relation:** is the number of attributes / columns of the relation. It is an **intrinsic** property of a relation. (Refer to structure)
- ❑ **Cardinality:** of a relation is the number of tuples / rows of the relation. It is **not an intrinsic** property of a relation, and represents a **state** of the relation. (refer to size)

ID	Name	Major
101	Vineet	CSE
102	Ayush	CSE
103	Surya	MTH
104	Ashish	MTH
105	Anshu	PHY
106	Sumit	CHY

Cardinality → 6

Degree/Arity → 3

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Relational Algebra

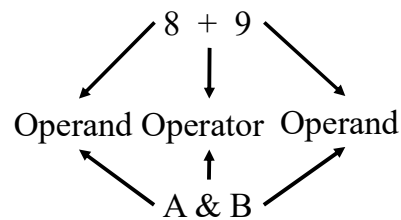


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Relational Algebra



- It is just like a simple operation; instead of number or variables as an operand we will have relation as operand).



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Operations



□ Basic/Fundamental Operations:

• **Unary Operations:**

- **Selection (σ):** Selects a subset of tuples / rows from a relation
- **Projection (π):** Retains a subset of columns from a relation
- **Rename (ρ):** Renames a relation

• **Binary operations:**

- **Cross Product / Cartesian Product ($\times$):** Allows to combine two relations
- **Difference / Set difference ($-$):** Returns tuples present in one relation but not in the other.
- **Union ($\cup$):** Consolidates tuples in two relations

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Operations



❑ Few More Operations:

- **Intersection ($\cap$):** Returns tuples present in both relations
- **Join ($\bowtie$):** Allows to combine two or more relations
 - There are several types of joins:
 - semi join, left outer join, right outer join, full outer join
- **Division ($/$):** Returns those tuples of a relation which subsets tuples of another table
- **Assignment ($\leftarrow$):** Assigning the output to a relation

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Operations: Selection (σ)



❑ Selection (σ):

- Selects a subset of tuple (rows) based on a given condition. (also known as **horizontal partitioning**)
- Syntax: $\sigma_{\langle \text{condition} \rangle}(\langle \text{Relation} \rangle)$
- $\sigma_{\text{Major}='CSE'}(\text{Students})$: Select all students majoring in CSE.

ID	Name	Major
101	Vineet	CSE
102	Ayush	CSE
103	Surya	MTH
104	Ashish	MTH
105	Anshu	PHY
106	Sumit	CHY



ID	Name	Major
101	Vineet	CSE
102	Ayush	CSE

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Selection (σ): Example 2



☐ Cascading conditions

☐ $\sigma_{\text{Major}='CSE' \wedge \text{Age}>19}(\text{Students})$

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE
103	Surya	18	MTH
104	Ashish	21	MTH
105	Anshu	22	PHY
106	Sumit	18	CHY



ID	Name	Age	Major
102	Ayush	20	CSE

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Selection (σ): Example 3



☐ Nested Conditions

☐ $\sigma_{\text{Major}='CSE'}(\sigma_{\text{Age}>19}(\text{Students}))$

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE
103	Surya	18	MTH
104	Ashish	21	MTH
105	Anshu	22	PHY
106	Sumit	18	CHY



ID	Name	Age	Major
102	Ayush	20	CSE
104	Ashish	21	MTH
105	Anshu	22	PHY



ID	Name	Age	Major
102	Ayush	20	CSE

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Selection (σ): Example 3

❑ Nested Conditions

❑ $\sigma_{\text{Major}='CSE'}(\sigma_{\text{Age}>19}(\text{Students}))$

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE
103	Surya	18	MTH
104	Ashish	21	MTH
105	Anshu	22	PHY
106	Sumit	18	CHY

ID	Name	Age	Major
102	Ayush	20	CSE
102	Ayush	20	CSE

ID	Name	Age	Major
102	Ayush	20	CSE

❑ Order of nesting does not affect the output → Select (σ) operator is **commutative**.

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Selection (σ) Operator Summary

- ❑ Selection operator works on a **single tuple at a time**
- ❑ The CONDITION involved in selection operation MUST be a Boolean expression (or a logical combination of more than one Boolean expressions).
- ❑ Selection operator may **reduce the cardinality** of a relation when it acts on, i.e.,
 $|\sigma_{\langle \text{CONDITION} \rangle}(\langle \text{RELATION} \rangle)| \leq |\langle \text{RELATION} \rangle|$
- ❑ Selection operator does not alter the arity of a relation
- ❑ Selection operator can act on a relation or any relational algebra expression resulting on a table.
- ❑ Selection operator is **commutative**.

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Projection (π): Example 1



- ❑ **Projection (π):** selects a subset of attributes from a relation.
Also called as *vertical partitioning*.
- ❑ **Syntax:** $\pi_{\langle \text{Attributes} \rangle}(\langle \text{Relation} \rangle)$
- ❑ $\pi_{\text{Name, Major}}(\text{Students})$

ID	Name	Age	Major		Name	Major
101	Vineet	19	CSE	➔	Vineet	CSE
102	Ayush	20	CSE		Ayush	CSE
103	Surya	18	MTH		Surya	MTH
104	Ashish	21	MTH		Ashish	MTH
105	Anshu	22	PHY		Anshu	PHY
106	Sumit	18	CHY		Sumit	CHY

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Projection (π): Example 2



- ❑ **Automatically removes duplicates.**
- ❑ $\pi_{\text{ID}}(\text{Enrolments})$

SN	ID	CourseID		ID
1	101	CS101	➔	101
2	102	CS102		102
3	103	MT101		103
4	102	EE101		104
5	104	CH102		105
6	105	CH101		

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Projection (π) Operator Summary



- ❑ *Is Projection (π) operation commutative?*
- ❑ We know, for a projection operation to work, the attributes it retrieves must be a subset of the attributes of the input relation.
 - For $\pi_{\langle \text{ATTRIBUTES1} \rangle}(\pi_{\langle \text{ATTRIBUTES2} \rangle}(\langle \text{RELATION} \rangle))$ to be valid, the following condition must be true:
 - $\langle \text{ATTRIBUTES2} \rangle \subseteq \langle \text{ATTRIBUTES1} \rangle$
- ❑ Also, for projection to be commutative,
 - $\pi_{\langle \text{ATTRIBUTES2} \rangle}(\pi_{\langle \text{ATTRIBUTES1} \rangle}(\langle \text{RELATION} \rangle)) = \pi_{\langle \text{ATTRIBUTES1} \rangle}(\pi_{\langle \text{ATTRIBUTES2} \rangle}(\langle \text{RELATION} \rangle))$
 - However,
 - $\pi_{\langle \text{ATTRIBUTES2} \rangle}(\pi_{\langle \text{ATTRIBUTES1} \rangle}(\langle \text{RELATION} \rangle)) \neq \pi_{\langle \text{ATTRIBUTES2} \rangle}(\langle \text{RELATION} \rangle)$
- ❑ **Hence projection is not commutative.**

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Projection (π) Operator Summary



- ❑ The $\langle \text{ATTRIBUTES} \rangle$ involved in projection operation MUST be a **subset** (may be equal) of **all attributes** in $\langle \text{RELATION} \rangle$
- ❑ Projection operator may **reduce the arity / degree** of a relation when it acts on.
- ❑ Projection operator **may alter the cardinality** of a relation as it removes duplicate tuples
- ❑ If $\langle \text{ATTRIBUTES} \rangle$ is a **superkey**, then $|\pi_{\langle \text{ATTRIBUTES} \rangle}(\langle \text{RELATION} \rangle)| = |\langle \text{RELATION} \rangle|$
- ❑ Projection operation defined in abstract relational algebra language eliminates duplicates, but projection operator in SQL retains duplicates.

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Cascading Selection and Projection

❑ $\pi_{\text{Name, Major}}(\sigma_{\text{Age} > 19}(\text{Students}))$

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE
103	Surya	18	MTH
104	Ashish	21	MTH
105	Anshu	22	PHY
106	Sumit	18	CHY



Name	Major
Ayush	CSE
Ashish	MTH
Anshu	PHY

- ❑ Please note that $\pi_{\text{Name, Major}}(\sigma_{\text{Age} > 19}(\text{Students})) \neq \sigma_{\text{Age} > 19}(\pi_{\text{Name, Major}}(\text{Students}))$,
- ❑ i.e., the roles of selection and projection cannot be commutative. Always selection acts first, and then projection.

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Rename (ρ)

❑ **Renaming relation name and attributes:**

- $\rho_{\langle \text{OUTPUT_RELATION} \rangle}(\langle \text{OUTPUT_ATTRIBUTES} \rangle)(\langle \text{INPUT_RELATION} \rangle)$

❑ **Renaming relation name only (and not attributes):**

- $\rho_{\langle \text{OUTPUT_RELATION} \rangle}(\langle \text{INPUT_RELATION} \rangle)$

❑ **Renaming attributes only (and not relation name):**

- $\rho(\langle \text{OUTPUT_ATTRIBUTES} \rangle)(\langle \text{INPUT_RELATION} \rangle)$

- ❑ The input for rename operator can be a relation or a relational algebra expression
- ❑ Rename operation is a special type of projection

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Set Operations



- ❑ The set operations **Union ($\cup$)**, **Intersection ($\cap$)**, **Difference ($-$)** are binary Operations.
- ❑ **Union ($\cup$)**: Consolidates tuples in two relations
- ❑ **Intersection ($\cap$)**: Extracts only common tuples from two relations
- ❑ **Difference ($-$)**: Extracts tuples present exclusively in first relation (but not present in second relation).

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Set Operations



- ❑ **Union Compatibility**: The two input relations should have (i) same degree and (ii) same domain for every attribute
- ❑ Union Compatibility is decided based on intrinsic properties of two relations, not based on it's state.
- ❑ $R \cup S$ automatically stores the **result in first operand, i.e., R** .
- ❑ **Union eliminates duplicates from the output and returns the common tuples only once in output, and intersection cannot have duplicates.**
- ❑ Intersection is a derived operation, i.e.
- ❑ $R \cap S = R - (R - S) = S - (S - R)$

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Set Operations



- ❑ Union and intersection operations are commutative, i.e.
 - $R \cup S = S \cup R$ and $R \cap S = S \cap R$
- ❑ Union and intersection operations are associative, i.e.
 - $(R \cup S) \cup T = R \cup (S \cup T)$ and $(R \cap S) \cap T = R \cap (S \cap T)$
- ❑ Difference is not commutative, i.e.,
 - $R - S \neq S - R$

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Cardinality Analysis of Set Operations



- ❑ Assume Relation A having cardinality m, B having cardinality n.

Operation ↓	Venn Diagram Representation ↓	Minimum Cardinality	Maximum Cardinality
$A \cup B$		$\max(m, n)$	$m+n$
$A \cap B$		0	$\min(m, n)$
$A - B$		0	m

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Union ($\cup$): Example



□ Students1 $\cup$ Students2

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE

$\cup$
↓

ID	Name	Age	Major
102	Ayush	20	CSE
103	Surya	18	MTH

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE
103	Surya	18	MTH

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Union ($\cup$): Example 2



□ $\pi_{ID}(\text{Students1}) \cup \pi_{ID}(\text{Students2})$

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE

$\cup$
↓

ID	Name	Age	Major
102	Ayush	20	CSE
103	Surya	18	MTH

ID
101
102
103

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Intersection ($\cap$): Example

- ❑ Returns the tuples that are common in two relations.
 - Students1 $\cap$ Students2

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE

$\cap$


ID	Name	Age	Major
102	Ayush	20	CSE

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Difference (-): Example

- ❑ Return tuple available in one relation but not in other.
 - Students1 - Students2

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE

$-$


ID	Name	Age	Major
101	Vineet	19	CSE

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Difference (-): Example 2

□ $\pi_{ID}(\text{Students1}) - \pi_{ID}(\text{Students2})$

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE

-

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE
103	Surya	18	MTH

No Output

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Cartesian Product (x)

- Cross product / Cartesian product ($\times$) is a binary Operation.
- The input relations need not be union compatible for participating in cross product.
 - Arity of $R \times S = (\text{Arity of } R + \text{Arity of } S)$
 - Cardinality of $R \times S = (\text{Cardinality of } R \times \text{Cardinality of } S)$

ID	Name	Major
101	Vineet	CSE
102	Ayush	CSE

$\times$

SN	CourseID
1	CS101
2	CS102

$\downarrow$

ID	Name	Major	SR	CourseID
101	Vineet	CSE	1	CS101
101	Vineet	CSE	2	CS102
102	Ayush	CSE	1	CS101
102	Ayush	CSE	2	CS102

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Cartesian Product ($\times$)



- ❑ Cross product is commutative, i.e., $R \times S = S \times R$
- ❑ Cross product is associative, i.e., $(R \times S) \times T = R \times (S \times T)$
- ❑ In most practical cases, such unconditional linking of every tuple of a relation with every tuple of another relation is *meaningless*.

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Join ($\bowtie$)



- Combines related tuple from two tables based on related column.
- Syntax: $P \leftarrow R \bowtie_{\langle \text{join condition} \rangle} S$
- Join product is commutative, i.e., $R \bowtie_{\langle \text{join condition} \rangle} S = S \bowtie_{\langle \text{join condition} \rangle} R$
- Arity of $R \bowtie S = (\text{Arity of } R + \text{Arity of } S)$
- Minimum Cardinality of $R \bowtie S = 0$
- Maximum Cardinality of $R \bowtie S = (\text{Cardinality of } R \times \text{Cardinality of } S)$
- $\langle \text{join condition} \rangle$ is denoted as $R_i \theta S_j$ where R_i and S_j are two attributes from the relations respectively, and θ is the **binary Boolean operation** between them.
- θ can be checking if they are **equal, or unequal, or one is greater than other** etc.
- As the operation between attributes from different relations is classically represented by θ , it is also called *theta join*.

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Join ($\bowtie$)

Types:

- **Natural Join:** Matches rows with the same value in common columns.
- **Theta Join:** Combines rows based on a condition.
- **Outer Join:** Includes unmatched rows (left, right, or full outer joins).

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Natural Join ($\bowtie$ or $*$)

- A special type of join operation, where $\langle \text{join condition} \rangle$ is not explicitly written.
- An equality check between the attribute pairs having same name is done, and accordingly the result of natural join is computed.
- $R \bowtie S$ or $R * S$

A	B	C		A
A1	B1	C1	*	A1
A2	B2	C2		A2
A3	B3	C2		

↓

A	B	C
A1	B1	C1
A2	B2	C2

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Natural Join ($\bowtie$ or $*$) : Example 2

Students $\bowtie$ Enrolments

StudentID	Name	Major		EnrollID	StudentID	CourseID
1	Vineet	CSE	*	101	1	CS101
2	Ayushman	CSE		102	2	CS102

↓

StudentID	Name	Major	EnrollID	CourseID
1	Vineet	CSE	101	CS101
2	Ayushman	CSE	102	CS102

Common attributes will not repeat in the resultant relation.

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Natural Join ($\bowtie$ or $*$) : Example 3

Students $\bowtie$ Enrolments

StudentID	Name	Major		EnrollID	StudentID	CourseID
1	Vineet	CSE	*	101	1	CS101
2	Ayushman	CSE		102	2	CS102
3	Ashish	MTH				

↓

StudentID	Name	Major	EnrollID	CourseID
1	Vineet	CSE	101	CS101
2	Ayushman	CSE	102	CS102

Common attributes will not repeat in the resultant relation.

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Theta Join ($\bowtie_{\theta}$)



- Combines rows based on a condition (θ).

$\text{Students} \bowtie_{\text{Students.StudentID} = \text{Enrollments.StudentID}} \text{Enrollments}$

StudentID	Name	Major
1	Vineet	CSE
2	Ayushman	CSE
3	Surya	MTH

EnrollID	StudentID	CourseID
1	1	CS101
2	3	CS102

$\bowtie_{\theta}$



StudentID	Name	Major	EnrollID	CourseID
1	Vineet	CSE	1	CS101
3	Surya	MTH	2	CS102

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Theta Join ($\bowtie_{\theta}$): Example 2



$\text{Students} \bowtie_{\text{Students.StudentID} > 1} \text{Enrollments}$

StudentID	Name	Major
1	Vineet	CSE
2	Ayushman	CSE
3	Surya	MTH

EnrollID	StudentID	CourseID
1	1	CS101
2	3	CS102

$\bowtie_{\theta}$



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Theta Join ($\bowtie_{\theta}$): Example 2

❑ $\text{Students} \bowtie_{\theta} (\text{Students.StudentID} > 1) \text{ Enrollments}$

- Apply the Condition $\text{StudentID} > 1$ on Enrollments.

EnrollID	StudentID	CourseID
1	1	CS101
2	3	CS102



EnrollID	StudentID	CourseID
2	3	CS102

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Theta Join ($\bowtie_{\theta}$): Example 2

❑ $\text{Students} \bowtie_{\theta} (\text{Students.StudentID} > 1) \text{ Enrollments}$

- Apply the Condition $\text{StudentID} > 1$ on Enrollments.
- Perform the Join: Combine filtered Enrollments relation with Students where $\text{Students.StudentID} = \text{Enrollments.StudentID}$.

EnrollID	StudentID	CourseID
2	3	CS102

$\bowtie_{\theta}$



StudentID	Name	Major
1	Vineet	CSE
2	Ayushman	CSE
3	Surya	MTH

StudentID	Name	Major	EnrollID	CourseID
3	Surya	MTH	2	CS102

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Inner Join



- Combines rows from two tables where there is a match based on a specified condition. It is a specific case of theta join where the condition is equality.

Students ⋈ (Students.StudentID = Enrollments.StudentID) Enrollments

StudentID	Name	Major		EnrollID	StudentID	CourseID
1	Vineet	CSE		1	1	CS101
2	Ayushman	CSE		2	3	CS102
3	Surya	MTH				

⋈

↓

StudentID	Name	Major	EnrollID	CourseID
1	Vineet	CSE	1	CS101
3	Surya	MTH	2	CS102

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Outer Join



- Combines rows from two tables, including unmatched rows, by filling in NULL values where matches are not found.
- Types of outer joins:
 - Left Outer Join
 - Right Outer Join
 - Full Outer Join

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Left Outer Join ($\bowtie$)



- ❑ Returns all rows from the left table (Students), and matching rows from the right table (Enrollments). Non-matching rows in the right table are filled with NULL.
- ❑ $\text{Students} \bowtie (\text{Students.StudentID} = \text{Enrollments.StudentID}) \text{Enrollments}$

StudentID	Name	Major	StudentID	CourseID
1	Vineet	CSE	1	CS101
2	Ayushman	CSE	4	CS103
3	Surya	MTH		

$\bowtie$

StudentID	Name	Major	StudentID	CourseID
1	Vineet	CSE	1	CS101
2	Ayushman	CSE	NULL	NULL
3	Surya	MTH	NULL	NULL

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Right Outer Join ($\bowtie$)



- ❑ Returns all rows from the right table (Enrollments), and matching rows from the left table (Students). Non-matching rows in the left table are filled with NULL.
- ❑ $\text{Students} \bowtie (\text{Students.StudentID} = \text{Enrollments.StudentID}) \text{Enrollments}$

StudentID	Name	Major	StudentID	CourseID
1	Vineet	CSE	1	CS101
2	Ayushman	CSE	4	CS103
3	Surya	MTH		

$\bowtie$

StudentID	Name	Major	StudentID	CourseID
1	Vineet	CSE	1	CS101
NULL	NULL	NULL	4	103

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Full Outer Join ($\bowtie$)



- Combine results of previous two joins.
- Students $\bowtie$ (Students.StudentID = Enrollments.StudentID) Enrollments

StudentID	Name	Major		StudentID	CourseID
1	Vineet	CSE		1	CS101
2	Ayushman	CSE		4	CS103
3	Surya	MTH			

$\bowtie$

StudentID	Name	Major	StudentID	CourseID
1	Vineet	CSE	1	CS101
2	Ayushman	CSE	NULL	NULL
3	Surya	MTH	NULL	NULL
NULL	NULL	NULL	4	103

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Semi Join ($\bowtie$)



- A semi-join between two relations returns tuples from the one relation where one or more matches (according to join condition) are found in tuples of the other relation.
- Though a tuple in one relation may have match with more than one tuples of the other relation, the tuple will appear only once in output.
- left semi-join ($R \ltimes S$) retains those tuples of R which have a match with S.
- right semi-join ($R \rtimes S$) retains those tuples of S which have a match with R
- Semi-join is not commutative.**

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Left Semi Join ($\bowtie$)



- ❑ Returns rows from the left table that have matching rows in the right table, without including columns from the right table.
- ❑ $\text{Students} \bowtie (\text{Students.StudentID} = \text{Enrollments.StudentID}) \text{Enrollments}$

StudentID	Name	Major		StudentID	CourseID
1	Vineet	CSE	$\bowtie$ ↓	1	CS101
2	Ayushman	CSE		3	CS102
3	Surya	MTH		4	CS103

StudentID	Name	Major
1	Vineet	CSE
3	Surya	MTH

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Right Semi Join ($\bowtie$)



- ❑ Returns rows from the right table that have matching rows in the left table, without including columns from the right table.
- ❑ $\text{Students} \bowtie (\text{Students.StudentID} = \text{Enrollments.StudentID}) \text{Enrollments}$

StudentID	Name	Major		StudentID	CourseID
1	Vineet	CSE	$\bowtie$ ↓	1	CS101
2	Ayushman	CSE		3	CS102
3	Surya	MTH		4	CS103

StudentID	Course ID
1	CS101
3	CS102

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Anti Join



- ❑ A anti-join or anti semi-join between two relations returns those tuples from the one relation for which there exists no corresponding matches (according to join condition) in tuples of the other relation.
- ❑ left anti-join or called ONLY anti-join ($R \triangleright S$) retains those tuples of R which do not have a match with S
 - $R \triangleright S = R - (R \bowtie S)$
- ❑ right anti-join ($R \triangleleft S$) retains those tuples of S which do not have a match with R.
 - $R \triangleleft S = S - (R \bowtie S)$
- ❑ *anti-join is not commutative.*

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Left Anti Join ($\triangleright$)



- ❑ Returns rows from the left table that do not have matching rows in the right table.
- ❑ $\text{Students} \triangleright \text{Enrolments} = \text{Students} - (\text{Students} \bowtie \text{Enrolments})$

StudentID	Name	Major		StudentID	CourseID
1	Vineet	CSE	$\triangleright$ 	1	CS101
2	Ayushman	CSE		3	CS102
3	Surya	MTH		4	CS103
StudentID	Name	Major			
2	Ayushman	CSE			

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Right Anti Join ($\Joinright$)



- ❑ Returns rows from the left table that do not have matching rows in the right table.
- ❑ $\text{Students} \Joinright \text{Enrollments} = \text{Enrollments} - (\text{Students} \Join \text{Enrollments})$
 $(\text{Students.StudentID} = \text{Enrollments.StudentID})$

StudentID	Name	Major	StudentID	CourseID
1	Vineet	CSE	1	CS101
2	Ayushman	CSE	3	CS102
3	Surya	MTH	4	CS103

$\Joinright$

StudentID	CourseID
4	CS103

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Cardinality Analysis of Join Operations



Operation ↓	Venn Diagram Representation ↓	Minimum Cardinality	Maximum Cardinality
$A \times B$	NO CONDITION	mn	mn
$A \bowtie B$		0	mn
$A \ltimes B$		0	m
$A \rtimes B$		0	n
$A \supset B$		0	m
$A \subset B$		0	n
$A \Join B$		m	mn
$A \Joinleft B$		n	mn
$A \Joinright B$		$\max(m, n)$	mn

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Division (/)

- ❑ $P \leftarrow R / S$ will retain those attributes which are present in R but not in S only for those tuples where ALL values of common attributes in R matches any tuple of S
- ❑ Syntax: $P \leftarrow R/S$
- ❑ Note: $P * S$ may not return R, Non-Commutative.
- ❑ Find students taken all required courses.

Name	CourseID		CourseID
Vineet	CS101	/	CS101
Vineet	CS102		CS102
Surya	CS103		
Ashish	CS102		

Name
Vineet

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Division (/)

- ❑ $P \leftarrow R / S$ will retain those attributes which are present in R but not in S only for those tuples where ALL values of common attributes in R matches any tuple of S
- ❑ Syntax: $P \leftarrow R/S$
- ❑ Find students taken all required courses.

Name	CourseID		CourseID
Vineet	CS101	/	CS101
Vineet	CS102		CS102
Surya	CS103		
Ashish	CS102		
Ashish	CS101		

Name
Vineet
Ashish

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Division (/)



- $T \leftarrow R / S$
 - $T1 \leftarrow \pi_{(R-S)}(R)$
 - $T2 \leftarrow \pi_{(R-S)}((S \times T1) - R)$
 - $T \leftarrow T1 - T2$

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Relational Calculus

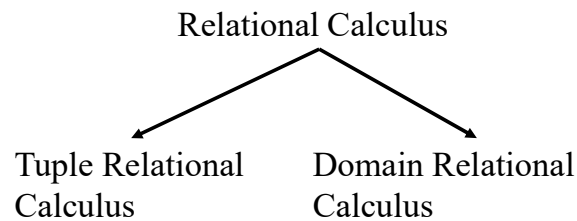


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Relational Calculus



- ❑ Relational calculus is a non-procedural query language used in relational databases to describe the desired results of a query without specifying how to compute them.
- ❑ It is based on mathematical logic and works with variables, predicates, and logical connectives to define queries.



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Relational Calculus



- ❑ TRC and DRC are equivalent in power.
- ❑ TRC and DRC are also equivalent in power with relational algebra.

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Tuple Relational Calculus



- ☐ Works with rows as variables
- ☐ Syntax: $\{t \mid P\{t\}\}$
 - t is a tuple
 - $P(t)$ is a predicted or condition that t must satisfy.
- ☐ **Operations:** $P(t)$ may have several conditions logically combined like AND ($\wedge$), OR ($\vee$), NOT ($\neg$), and quantifiers like EXISTS ($\exists$) and FOR ALL ($\forall$).
- ☐ **Atomic Operation:** if we use single operation.
- ☐ **Composite:** Multiple operations.

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Tuple Relational Calculus



- ☐ Name of the Students.
 - $\{Students.Name \mid Students(s)\}$

ID	Name	Age	Major		Name
101	Vineet	19	CSE	➔	Vineet
102	Ayush	20	CSE		Ayush
103	Surya	18	MTH		Surya
104	Ashish	21	MTH		Ashish

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Tuple Relational Calculus



□ Name of the Students enrolled in CSE.

- {Students.Name | Students (s)[^]s.Major = 'CSE'}

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE
103	Surya	18	MTH
104	Ashish	21	MTH



Name
Vineet
Ayush

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Tuple Relational Calculus



□ Name of the Students enrolled in CSE and Age>20.

- {Students.Name | Students (s)[^]s.Major = 'CSE'[^]s.Age>19}

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE
103	Surya	18	MTH
104	Ashish	21	MTH



Name
Vineet
Ayush

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Tuple Relational Calculus

□ Name of the Students enrolled in CSE and Age > 20.

- $\{ \text{Students.Name} \mid \text{Students}(s) \wedge (\exists e) (\text{Enrolments}(e) \wedge s.\text{ID} = e.\text{StudentID} \wedge \text{CourseID} = \text{'CS101'}) \}$

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE
103	Surya	18	MTH
104	Ashish	21	MTH



StudentID	CourseID
101	CS101
103	CS102
104	CS103

Name
Vineet

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Domain Relational Calculus

□ Works with domain variables (individual attributes/fields).

□ Syntax: $\{x_1, x_2, \dots, x_n \mid P(x_1, x_2, \dots, x_n)\}$

- $x_1, x_2, \dots$ are domain variables
- $P(x_1, x_2)$ is a predicate or condition that the domain variable must satisfy.
- i.e. Name of the students enrolled in CSE.
 - $\{ \langle x \rangle \mid \exists y (x, y) \in \text{Students} \wedge y = \text{'CSE'} \}$

ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE
103	Surya	18	MTH
104	Ashish	21	MTH



ID	Name	Age	Major
101	Vineet	19	CSE
102	Ayush	20	CSE

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GATE Questions

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Gate Questions

Q. Which of the following query transformations (i.e. replacing the l.h.s. expression by the r.h.s. expression) are incorrect?
 R_1 and R_2 are relations, C_1 and C_2 are selection conditions, and A_1 and A_2 are attributes of R_1 .

- (a) $\sigma_{C_1}(\sigma_{C_2}(R_1)) \rightarrow \sigma_{C_2}(\sigma_{C_1}(R_1))$
- (b) $\sigma_{C_1}(\pi_{A_1}(R_1)) \rightarrow \pi_{A_1}(\sigma_{C_1}(R_1))$
- (c) $\sigma_{C_1}(R_1 \cup R_2) \rightarrow \sigma_{C_1}(R_1) \cup \sigma_{C_1}(R_2)$
- (d) $\pi_{A_2}(\sigma_{C_1}(R_1)) \rightarrow \sigma_{C_1}(\pi_{A_2}(R_1))$

[GATE1998]

- (a) Correct: As the selection operator is commutative, the order of applying it will not impact the result.
- (b) Incorrect: Projection & Selection combined is non commutative.
- (c) correct: Selection distributed over union both are commutative.
- (d) Incorrect: In RHS if selection operator requires attribute A_2 , then the transformation will become invalid.

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Gate Questions



Q. Suppose $R_1(A, B)$ and $R_2(C, D)$ are two relation schemas. Let r_1 and r_2 be the corresponding relation instances. B is a foreign key that refers to C in R_2 . If the data in r_1 and r_2 satisfy referential integrity constraints, which of the following is ALWAYS TRUE?

- (a) $\pi_B(r_1) - \pi_C(r_2) = \phi$
- (b) $\pi_C(r_2) - \pi_B(r_1) = \phi$
- (c) $\pi_B(r_1) = \pi_C(r_2)$
- (d) $\pi_B(r_1) - \pi_C(r_2) \neq \phi$

[GATE2012]

- (a) Always True: as B is foreign key, it will be present in R_1 and R_2 both.
- (b) Not always true: C may be present in R_2 , but not necessarily in R_1 .
- (c) Not always true: again C may contain additional values not referenced by R_1 .
- (d) Always False: not possible as B is foreign key, and the relation satisfy referential integrity constraints.

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Gate Questions



Q. Give a relational algebra expression using only the minimum number of operators from $\{\cup, -\}$ which is equivalent to $R \cap S$.

[GATE1994]

- $R - R(R-S)$:
- $R-S$: computes the set of tuples in R but not in S
- $R-(R-S)$, removes from R all tuples that are not in S , leaving only the tuples common to both R and S . (reason for minimality: because it uses only $(-)$ operator.
- Other non minimal solutions:
 - $R \cap S = (R \cup S) - ((R-S) \cup (S-R))$
 - $(R - (R-S)) - (S-R)$

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Gate Questions



Q. Consider a join of a relation R with a relation S. If R has m tuples and S has n tuples, then the maximum and minimum sizes of the join respectively are:

- (a) $m+n$ and 0
 - (b) mn and 0
 - (c) $m+n$ and $|m-n|$
 - (d) mn and $m+n$
- [GATE1999]

- Max: mn (every tuple of r matches with every tuple in S)
- Min: 0 (no match).

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Gate Questions



Q. Consider the following relation $P(X, Y, Z)$, $Q(X, Y, T)$ and $R(Y, V)$:
How many tuples will be returned by the following relational algebra query?

$$\pi_x(\sigma_{(P.Y=R.Y \wedge R.V=V(P \times R)))} - \pi_x(\sigma_{(Q.Y=R.Y \wedge Q.T>2(Q \times R))})$$

- (a) 3
 - (b) 1
 - (c) 2
 - (d) 4
- [GATE2019]

Ans: Option (b)

P			Q			R	
X	Y	Z	X	Y	T	Y	V
X1	Y1	Z1	X2	Y1	2	Y1	V1
X1	Y1	Z2	X1	Y2	5	Y3	V2
X2	Y2	Z2	X1	Y1	6	Y2	V3
X2	Y4	Z4	X3	Y3	1	Y2	V2

- First Part:
 - Perform Cartesian product $P \times R$
 - Select $P.Y = R.Y$ AND $R.V = V$
 - Project attribute X from result $\rightarrow X1, X2$
- Second Term:
 - Perform $Q \times R$.
 - Select $Q.Y = R.Y$ AND $Q.T > 2$
 - Project attribute X from result $\rightarrow X2$
- At last compute difference: X retrieved in first term – X retrieved in second term. Result $\rightarrow X1$

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